

Vidit Jain

☎ +971 555458369 • ✉ f20240363@dubai.bits-pilani.ac.in • in viditjain7

Computer Science undergraduate at BITS Pilani Dubai Campus, working at the intersection of Quantum Computing, Machine Learning, and Algorithm Design. Currently researching defenses against circuit-level threats in Quantum Federated Learning, with broader interests in quantum algorithms, reliable computing, and learning systems.

Education

2024–Present: B.E. Computer Science, Birla Institute of Technology and Science, Pilani – Dubai
Expected Graduation: 2028 | Current CGPA: 9.45
Planned Minor in Quantum Computing (Spring 2027)

Research Experience

09/2026–Present: Defense Against Circuit-Level Threats in Quantum Federated Learning
Advisor: Dr. Ashish Gupta, BITS Pilani – Dubai Campus

- Developing countermeasures against CULT attacks that inject unitary operators (e.g., Pauli-rotations, phase-kickback sign-flips) into parameterized quantum circuits.
- Analyzing state deviations and measurement expectation shifts to mitigate per-sample gradient deviations during local optimization.
- Formulating robust aggregation strategies to limit global model drift under attacker stealth-budget constraints.

02/2026–10/2026: UAE Migration Nowcasting | Big Data Analytics

Advisor: Dr. Mrutyanjaya Sahu, BITS Pilani – Dubai Campus

- Combined Facebook Marketing API and FCSC data using Bayesian hierarchical modeling to nowcast migration stocks.
- Calibrated models on 2015–2019 data to correct demographic bias in digital traces.
- Produced high-frequency forecasts for urban planning.

02/2026–10/2026: Counterfactual Explanations for Dubai Public Transportation

Advisor: Dr. Mrutyanjaya Sahu, BITS Pilani – Dubai Campus

- Developed counterfactual explanations with NSGA-II to identify minimal infrastructure interventions for predicted transport bottlenecks.
- Used GRU/LSTM models on Dubai RTA time-series data to forecast short-term ridership and station congestion.
- Translated model outputs into actionable infrastructure recommendations.

02/2025–06/2025: BCH+ ML Hybrid ECC Model

Independent Project

- Designed a BCH + ML error-correction model for multi-bit errors in high-speed RAM.
- Evaluated error reduction and latency for high-performance computing.

Notable Projects

GitHub: VIDITJAIN7

Fractal | AI-Powered Semantic Search Engine

- Built a hybrid search engine using Reciprocal Rank Fusion (RRF) for keyword and semantic results.
- Reduced hallucinations and achieved sub-200ms latency through model calibration and uncertainty estimation.

Data Poisoning Detection Pipeline

- Built an end-to-end pipeline to inject and detect label-flip and BadNets-style backdoor attacks in CIFAR-10 using ResNet-18.
- Implemented detection filters including Spectral Signatures (SVD) and Activation Clustering (PCA + KMeans), mitigating backdoor Attack Success Rate (ASR) to 1.22%.
- Optimized training throughput using mixed precision (AMP) and TF32 operations on NVIDIA RTX GPUs.

AeroTrack AI | Predictive Maintenance Platform

- Built a custom Fivetran Connector for real-time flight data ingestion into Google BigQuery.
- Trained Vertex AI AutoML models for maintenance prediction and fault detection.
- Added a Gemini interface for fleet managers to query aircraft status.

Questly | Gamified Engagement & Learning Platform

- Developed a full-stack platform designed to enhance user engagement through gamified "quests" and real-time feedback loops.
- Implemented a responsive front-end architecture using Next.js and Tailwind CSS, ensuring seamless cross-device compatibility.
- Engineered real-time state management for leaderboards and quest progress, optimizing for low-latency user interactions.

Work Experience

09/2025–12/2025: Professional Assistant, BITS Pilani – Dubai

- Assisted faculty in the C Programming Lab.
- Supported and mentored junior students in programming fundamentals and system design.

Technical Skills

Quantum Computing: Qiskit, PennyLane

Algorithms & Systems: Data Structures & Algorithms, Digital Logic, Vector Databases

AI/Machine Learning: Vertex AI (AutoML, Gemini), Semantic Search, LLM Calibration, Uncertainty Estimation

Languages: Python (PyTorch, TensorFlow), Java (Data Structures, Swing), Verilog, SQL, JavaScript

Cloud & Tools: Google Cloud Platform (BigQuery, Vertex AI), Fivetran, Git

Cybersecurity: Network Traffic Analysis (Wireshark), Penetration Testing (Kali Linux), Memory Corruption

Honors & Awards

- 3rd Place: Ready Player One Hackathon (Microsoft Tech Club, Dubai).
- 13th Place Overall: RIFT CTF 2025 (GDG BITS Dubai).
- Rank 14: Cyberverse CTF 2024 (Exploit3rs & GDG BITS Dubai).
- Global Rank 1294 (Top 5%): Cyber Apocalypse CTF 2025 (Hack The Box)
- Global Rank 1044: Holmes CTF (Hack The Box).